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ODD Protocol

Bayesian agent-based model of infectious disease transmission in an urban center

I. Overview

1. Purpose of the model

This SEIRD *(susceptible – Exposed – Infected – Recovered – Dieseased)* simulates the spread of an infectious agent through a heterogeneous urban environment structured around three spatially distinct zones: a dense urban core where agents work, a lower-density suburban, residential periphery, and one very densely populated suburban periphery mirroring European cities' segregated poorer neighborhoods. Agents are individuals and the health ministry of a fictional country over a complete year. Seasonality is introduced as a modifier of how fast the epidemy spreads, to mirror the contagious profile of the seasonal flu. The investigation explores the two following key policy questions:

- *How do individual preferences shape the transmission rate and death toll of an infectious disease in a heterogeneous urban environment?*
- *How does government entities' signaling shape transmission rates of the disease, through availability of information and level of trust in institutions?*

2. Entities, state variables and scales

2.1 Entities and their state variables

The model contains two classes of agents: individuals, and the health ministry. The table below lists their respective state variables

Individuals	Health Ministry
Susceptible (not infected)	No alert
Incubating	Alert state 1 (lowly coercive measures)
Infectious asymptomatic	Alert state 2 (medium coercive measures)
Infectious symptomatic	Alert state 3 (highly coercive measures)
Recovered (immune for 6 months)	
Dead	

In addition to epidemic state, each agent carries the following individual attributes:

- **residence:** residential zone assigned at initialization and fixed throughout
- **work_place:** work zone assigned at initialization and fixed throughout
- **age_group:** affects fatality rate
- **transport_pref:** preferred mode when commuting to work
- **sensibility_plan:** individual responsiveness to the public health communication campaign; drawn once at initialization from a residence-level prior
- **doctor:** a binary variable capturing whether the individual visits a doctor when they are sick with symptoms. The doctor always reports to the health ministry.

2.3 Environment

The spatial environment is a 100 × 100 grid partitioned into three zones. The urban core occupies the central 40 × 40 cells; a dense periphery occupies a 15 x 20 cells; the less dense periphery occupies the rest of the environment. Each cell can host at most 5 agents; contacts occur among co-located agents and their immediate neighborhood (radius r = 1).

2.4 Scales

- Temporal resolution: 1 tick = 12 hours over 365 days (730 ticks)
- Spatial resolution: 100 × 100 cells
- Population size: N = 5 000
- Observation schedule: aggregate cases are updated every 2 ticks (daily) for comparison between real prevalence and surveillance data

3. Process Overview and Scheduling

At initialization, agents' individual characteristics and preferences are drawn stochastically. Each morning (the first tick, then every other tick), healthy and asymptomatic individuals go to work. Individuals sick with symptoms decide whether to go to work, stay home, or stay home and go to the doctor (going to the doctor happens once per individual and per infection cycle). Visiting a doctor updates the count of infected individuals every day, which triggers a government response at certain thresholds. Each evening, individuals at work commute back home using the same mode of transportation as in the morning.

Every day, the government gets a report on the epidemic situation as a percentage of infected population over the last 7 days. In response, at certain thresholds, the government advertises containment measures, based on which individuals adapt their behavior. Containment plans cannot be downgraded before two weeks after implementation. Infections happen during commute to and from work, at work or at home (during the day) or during commute and at home (during the night), and depend on the infectiousness of disease, number of individuals within radius at work and at home, and on preferred mode of transportation.

II. Design Concepts

4.1 Basic principles

The infection part is based on a SEIRD (susceptible, exposed, infected, recovered, dead). The basic principle says that when one susceptible agent is within radius of an infected one, it gets infected with some probability. Transmission is driven by co-location and mobility; responses are shaped by individual preferences and interventions of the Public Health Ministry.

4.2 Emergence

The emergence in the model is the way the epidemic is spreading – the timing and strength of epidemic waves and the activation of public health interventions driven by agent behavior.

4.3 Adaptation

Agents adapt their behaviors based on the current state of epidemics and the policy signals set by Health Ministry. When the plan is active, individuals reduce the probability of commuting.

4.4 Sensing

Agents that are contagious but asymptomatic are not aware of their possibility to infect other, while the infectious symptomatic are fully aware and can predict that they may infect others.

4.5 Interaction

There are direct interactions between agents who are located in the same cell or in the neighboring cell. The interactions determine the probability of disease transmission.

4.6 Stochasticity

Stochasticity in the model exists during infection events, state transitions, mobility decisions, and medical consultations. Random draws are implemented using Bernoulli trials.

4.7 Collectives

Individuals form two groups according to where they live. Those who live in the denser periphery have lower sensibility to the government's containment measures, to reflect asymmetry of information between socio-economic segments of society.

4.8 Observations

Data is collected on the number of agents that are newly infected, susceptible, exposed, infected, recovered and dead at every time step. Model outputs also include peak incidence and timing of the epidemic peak

III. Details

5. Initialization

At the beginning of the simulation, individuals are created and assigned attributes. They are stochastically distributed across the two zones where they live - periphery and dense periphery and one where they work - center. The initialization also stochastically attaches information about age group, transport preference, responsiveness to the public health campaigns and the likelihood of visiting a doctor in case of infection.

Initial epidemic conditions are set by assigning a part of the agents to the exposed or infectious states. The majority of agents start in a susceptible state, and no agents are initially recovered. At the beginning, there is no active health recovery plan.

6. Input data

All parameters are fixed prior to simulation runs, which includes: epidemiological parameters, demographic structure and mobility patterns. There are no real-time data streams used. All variability arises from initial stochasticity and agent interactions.

7. Submodels

7.1 Mobility submodel

At each step, agents make a decision whether to commute based on their epidemic state and status of the health plan. People with symptoms and the ones responsive to the government health plan are less likely to travel to work.

7.2 Transmission submodel

The likelihood of transmission is computed from infectious neighbors incorporating: transmission rate, transport multiplier, and agent individual compliance.

7.3 Disease progression submodel

Transitions of agents between epidemiological states follow stochastic processes driven by agent interactions. Exposed individuals are contagious and become infected (asymptomatic or symptomatic). Eventually, they recover or die according to the severity of the disease and age group of the individual.

The transmission rate of the disease is highest in Winter, like the seasonal flu in European countries.

7.4 Medical consultation and reporting submodel

On the first morning of being symptomatic, individuals decide whether they want to see a doctor. Doctors always report to the government.

7.5 Health policy submodel

The health ministry activates health policy plans after reaching the threshold of θ infected individuals. They reduce the transmission of the virus as agents adapt their behavior in response to activation.